hotel_reservation.sql

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The Hotel Reservation System is a database project that manages guest details, room information, bookings, and payments. It stores and organizes data efficiently using tables and relationships. The system performs operations like booking rooms, checking availability, tracking reservations, and calculating charges, ensuring smooth and accurate hotel management operations.

```
CREATE DATABASE hotel_db;

USE hotel_db;


CREATE TABLE Guests (

   guest_id INT PRIMARY KEY AUTO_INCREMENT,

   name VARCHAR(100) NOT NULL,

   email VARCHAR(100) UNIQUE,

   phone VARCHAR(15)

);


CREATE TABLE Rooms (

   room_id INT PRIMARY KEY AUTO_INCREMENT,

   room_type VARCHAR(50),

   price_per_night DECIMAL(10,2),

   status VARCHAR(20)

);


CREATE TABLE Reservations (

   reservation_id INT PRIMARY KEY AUTO_INCREMENT,

   guest_id INT,

   room_id INT,

   check_in DATE,

   check_out DATE,

   status VARCHAR(20),
```

```
    FOREIGN KEY (guest_id) REFERENCES Guests(guest_id),
    FOREIGN KEY (room_id) REFERENCES Rooms(room_id)
);

CREATE TABLE Payments (
    payment_id INT PRIMARY KEY AUTO_INCREMENT,
    reservation_id INT,
    amount DECIMAL(10,2),
    payment_date DATE,
    FOREIGN KEY (reservation_id) REFERENCES Reservations(reservation_id)
);

INSERT INTO Guests (name, email, phone) VALUES
('Vanshika Dixit', 'vanshika@gmail.com', '9876543210'),
('Roshani Singh', 'roshani@gmail.com', '9123456780'),
('Any friend', 'amit@gmail.com', '9988776655');

INSERT INTO Rooms (room_type, price_per_night, status) VALUES
('Deluxe', 3000, 'Available'),
('Standard', 2000, 'Available'),
('Suite', 5000, 'Available');

INSERT INTO Reservations (guest_id, room_id, check_in, check_out, status) VALUES
(1, 1, '2026-04-01', '2026-04-05', 'Booked'),
(2, 2, '2026-04-02', '2026-04-04', 'Booked');

INSERT INTO Payments (reservation_id, amount, payment_date) VALUES
(1, 12000, '2026-04-01'),
(2, 4000, '2026-04-02');

SELECT * FROM Rooms
```

```
WHERE room_id NOT IN (
    SELECT room_id FROM Reservations
    WHERE status = 'Booked'
);
```

```
SELECT g.name, r.room_id, r.check_in, r.check_out, r.status
FROM Reservations r
JOIN Guests g ON r.guest_id = g.guest_id;
```

Result Grid | Filter Rows: | Edit:

room_id	room_type	price_per_night	status
3	Suite	5000.00	Available
NULL	NULL	NULL	NULL

```
SELECT rm.room_id, g.name
FROM Reservations r
JOIN Rooms rm ON r.room_id = rm.room_id
JOIN Guests g ON r.guest_id = g.guest_id
WHERE r.status = 'Booked';
```

Result Grid | Filter Rows: | Export:

room_id	name
1	Vanshika Dixit
2	Roshani Singh

```
UPDATE Reservations
```

```
SET status = 'Cancelled'
WHERE reservation_id = 1;

SELECT * FROM Guests;
SELECT * FROM Rooms;
SELECT * FROM Reservations;
```

Result Grid | Filter Rows: | Edit: | Export/Im

reservation_id	guest_id	room_id	check_in	check_out	status
1	1	1	2026-04-01	2026-04-05	Cancelled
2	2	2	2026-04-02	2026-04-04	Booked
NULL	NULL	NULL	NULL	NULL	NULL